

Complete checklist to perform an in-place upgrade of SQL Server 2012 Enterprise Edition to SQL Server 2019 Enterprise Edition in a Failover Cluster Instance (FCI) environment—without Always On Availability Groups (AGs), but with Windows Server Failover Clustering (WSFC). This checklist includes all critical steps and a **rollback plan in detail**, tailored specifically for **SQL Server FCIs**.

✅ SQL Server In-Place Upgrade Checklist (2012 → 2019 Enterprise) in Failover Cluster Environments

📌 1. PRE-UPGRADE PLANNING

✅ 1.1 Review Compatibility and Support

- Confirm SQL Server 2012 → 2019 in-place upgrade is supported for your OS and FCI setup.
- Review [Microsoft Supported Upgrade Paths](#) (2012 → 2019 is supported).
- Check that your **Windows Server OS** is supported by SQL Server 2019 (e.g., Windows Server 2016 or later).
- Verify the **cluster is healthy** (Validate Cluster in Failover Cluster Manager).

✅ 1.2 Review Feature Deprecations

- Use **Data Migration Assistant (DMA)** to identify deprecated or unsupported features.
- Run **SQL Server Upgrade Advisor** for SQL 2019.

✅ 1.3 Prepare Testing Environment

- Create a **lab/test cluster** that mirrors the production FCI for dry-run upgrade.
- Test upgrade steps and rollback process in this environment.

📁 2. BACKUP AND DOCUMENTATION

✅ 2.1 Backup and Export

- Full backups of all **user and system databases** (master, msdb, model).
- Backup SQL Agent jobs, SSIS packages, linked servers, credentials, etc.
- Export configurations: server-level settings, cluster settings, registry keys if needed.

✅ 2.2 Snapshot and Recovery Options

- Take **full VM or disk snapshots** of all cluster nodes before upgrade.
- Ensure **OS and SQL binaries are backed up**.
- Ensure storage-level **volume shadow copies (VSS)** or SAN-level snapshots are available.

🔧 3. PRE-UPGRADE VALIDATION

- Run DBCC CHECKDB on all databases.
- Validate SQL Server configuration settings (sp_configure).
- Document all installed **SQL Server components and features**.
- Check that **cluster resource group** for SQL Server is online and healthy.
- Verify disk space (minimum 20–30 GB free on system and data drives).
- Make sure **no pending Windows updates** are waiting to reboot.

4. UPGRADE EXECUTION (ROLLING NODE UPGRADE)

4.1 FCI Node Overview

In FCI, multiple nodes share one SQL Server instance with shared storage. The upgrade must be **performed on the passive node first**, followed by a **failover**, and then upgrade the second node.

4.2 Upgrade Steps

Step 1: Passive Node Upgrade

- Identify the **active and passive nodes** in the cluster.
- On the passive node:
 - Remove SQL Server instance from this node (using **Remove Node** setup option).
 - Reboot node.
 - Install SQL Server 2019 using the **Add Node** wizard and point to the existing FCI.
 - Reboot after installation.

Step 2: Failover

- Perform **planned failover** to the newly upgraded node using **Failover Cluster Manager** or Move-ClusterGroup.
- Validate SQL services and application connections.
- Run health checks and smoke tests.

Step 3: Upgrade Second Node

- Repeat the same process for the second (original active) node:
 - Remove SQL Server 2012 node.
 - Reboot.
 - Install SQL Server 2019 and join the existing cluster.
 - Reboot again.

5. POST-UPGRADE STEPS

- Verify **SQL Server version** using SELECT @@VERSION.
- Run DBCC CHECKDB on all databases post-upgrade.
- Reconfigure and validate:
 - SQL Server Agent jobs
 - Linked Servers
 - Maintenance Plans
 - Logins
 - Security settings
- Update database **compatibility level** to 150 (SQL Server 2019) after testing.
- Rebuild all indexes and update statistics.
- Review and update backup strategies.
- Test failover between upgraded nodes.
- Enable monitoring and alerts.

6. ROLLBACK PLAN – IN DETAIL

Since in-place upgrades are **destructive** (i.e., once the database engine is upgraded, it can't be downgraded), rollback requires a full **system-level restore** or use of **snapshots/standby**.

6.1 Rollback Triggers

- Installation fails or errors during upgrade.
- Application failure or performance degradation post-upgrade.
- Database corruption or unexpected behavior.

6.2 Rollback Strategies

A If SQL Services Were Not Upgraded Yet

- Rollback is simple—just power off the passive node, restore VM snapshot or uninstall SQL 2019.
- Failover back to original active SQL 2012 node.

B If SQL Server 2019 Has Taken Control of the Cluster

Rollback is more complex and involves full recovery:

1. **Power Down All Upgraded Nodes**
 - Isolate the upgraded SQL Server nodes.
2. **Restore VM Snapshots**
 - Restore all cluster nodes to pre-upgrade snapshot state (SQL 2012 binaries, registry, configs).
3. **Restore Shared Storage**
 - Ensure the **shared disks** used for the SQL instance are reverted to the pre-upgrade state.
4. **Recover Cluster Configuration**
 - Use cluster backups (PowerShell or manually backed up) to restore cluster state.
5. **Restore Databases (if needed)**
 - If database schema has been altered during the upgrade, restore full DB backups to SQL 2012.
6. **Test Application Connectivity**
 - Confirm application can again connect and run against the restored SQL Server 2012.

Optional Enhancements

- Script the **cluster configuration** (Get-Cluster, Get-ClusterGroup, etc.).
- Review and document **SQL Server logs** (ERRORLOG) and **Windows Event Logs**.
- Ensure **latest SQL Server 2019 CU** is applied after upgrade.
- Test **HA and DR scenarios** (failover, disk loss, quorum loss).

Summary Checklist

Phase	Tasks
Pre-Upgrade	Compatibility check, backups, DMA run, cluster health validation
Upgrade Step 1	Remove passive node → upgrade → rejoin
Upgrade Step 2	Failover to upgraded node
Upgrade Step 3	Remove and upgrade remaining node → rejoin cluster
Post-Upgrade	Reconfigure, update stats/indexes, compatibility levels, monitoring
Rollback Plan	VM snapshots, cluster backups, database restores if needed

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